



US 20250287129A1

(19) **United States**

(12) **Patent Application Publication**
HOCKMAN et al.

(10) **Pub. No.: US 2025/0287129 A1**

(43) **Pub. Date: Sep. 11, 2025**

(54) **INTEGRATED AUDIOVISUAL SYSTEM**

H04R 1/40 (2006.01)

(71) Applicant: **Sphere Entertainment Group, LLC,**
New York, NY (US)

H04R 3/12 (2006.01)

(72) Inventors: **Erik HOCKMAN**, New York, NY
(US); **Alex LUTHWAITE**, Lancaster,
PA (US); **Emad YACoub HANNA**,
Berlin (DE)

(52) **U.S. Cl.**

CPC **H04R 1/028** (2013.01); **G06F 3/147**
(2013.01); **H04R 1/403** (2013.01); **H04R 3/12**
(2013.01); **H04R 2201/403** (2013.01); **H04R**
2499/15 (2013.01)

(21) Appl. No.: **19/218,870**

(57)

ABSTRACT

(22) Filed: **May 27, 2025**

Related U.S. Application Data

(63) Continuation of application No. 18/521,021, filed on
Nov. 28, 2023, now Pat. No. 12,348,918, which is a
continuation of application No. 17/734,995, filed on
May 2, 2022, now Pat. No. 11,856,346, which is a
continuation of application No. 16/898,142, filed on
Jun. 10, 2020, now Pat. No. 11,375,305.

(60) Provisional application No. 62/859,930, filed on Jun.
11, 2019.

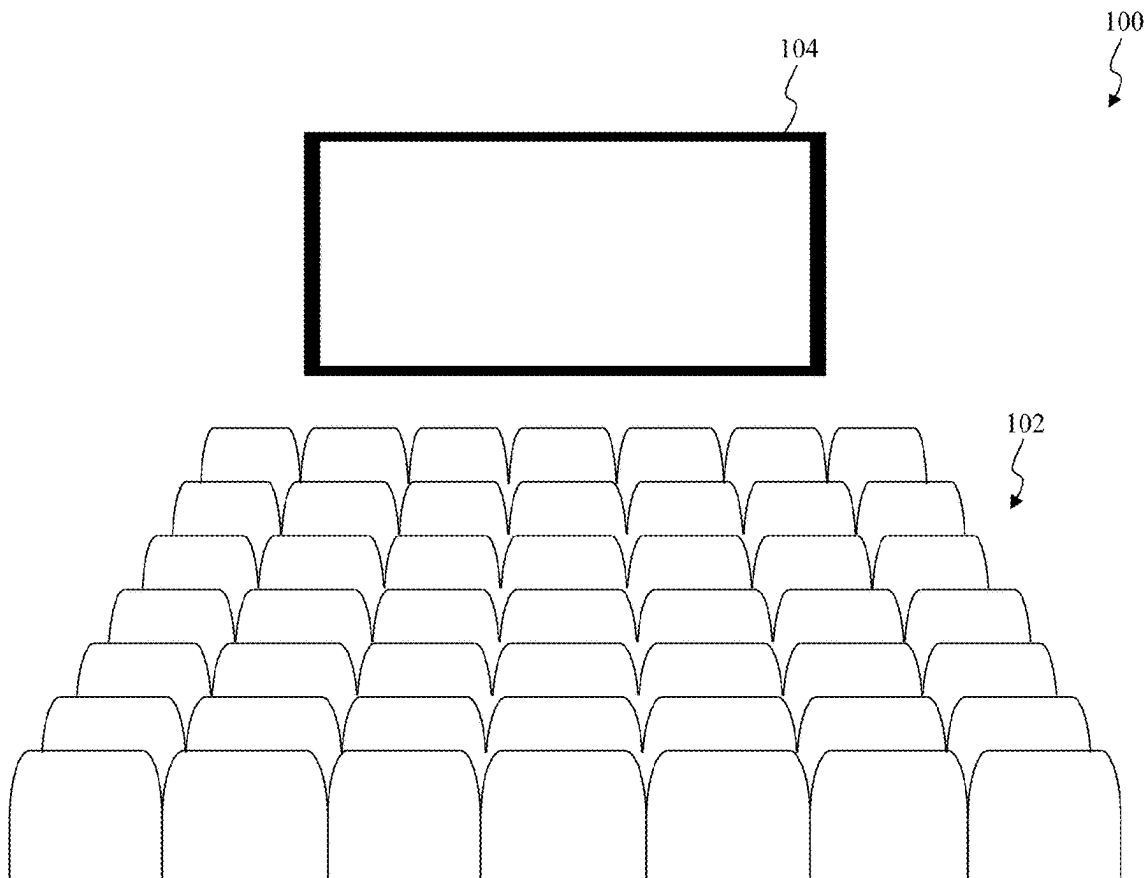
Publication Classification

(51) **Int. Cl.**

H04R 1/02 (2006.01)

G06F 3/147 (2006.01)

An integrated audio-visual system is disclosed for delivering an event to one or more spectators. The integrated audio-visual system includes one or more loudspeakers that are positioned behind one or more visual displays to be effectively behind the one or more visual displays to be shielded from the field of view of the one or more spectators. The one or more visual displays are specially designed and manufactured to allow sound associated with the event to propagate from the one or more loudspeakers with minimal acoustical distortion and/or minimum acoustical vibration while presenting a visual representation of the event to the one or more spectators. Moreover, the one or more loudspeakers and the one or more visual displays are situated to be a predetermined displacement from each other to further minimize acoustical distortion.



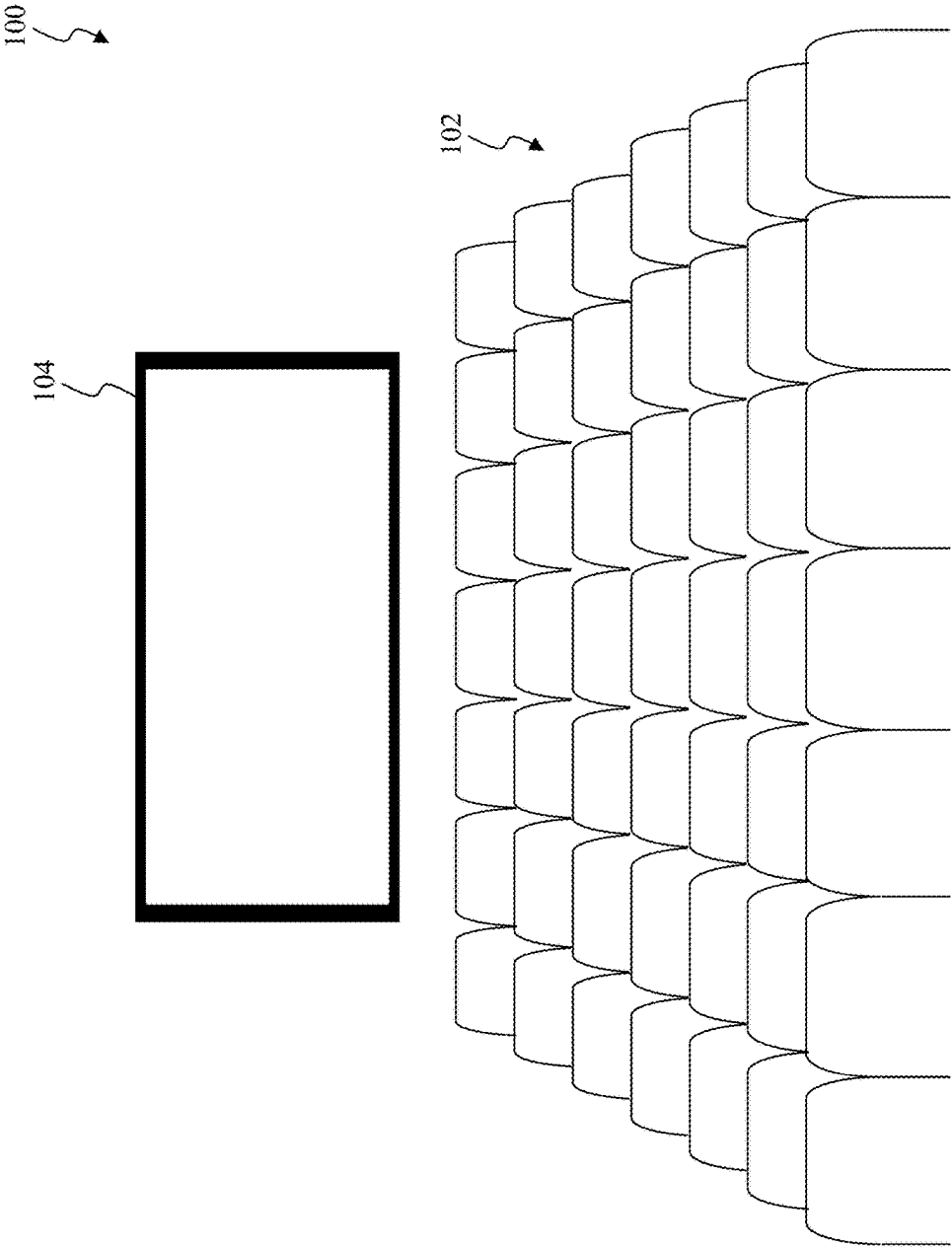


FIG. 1

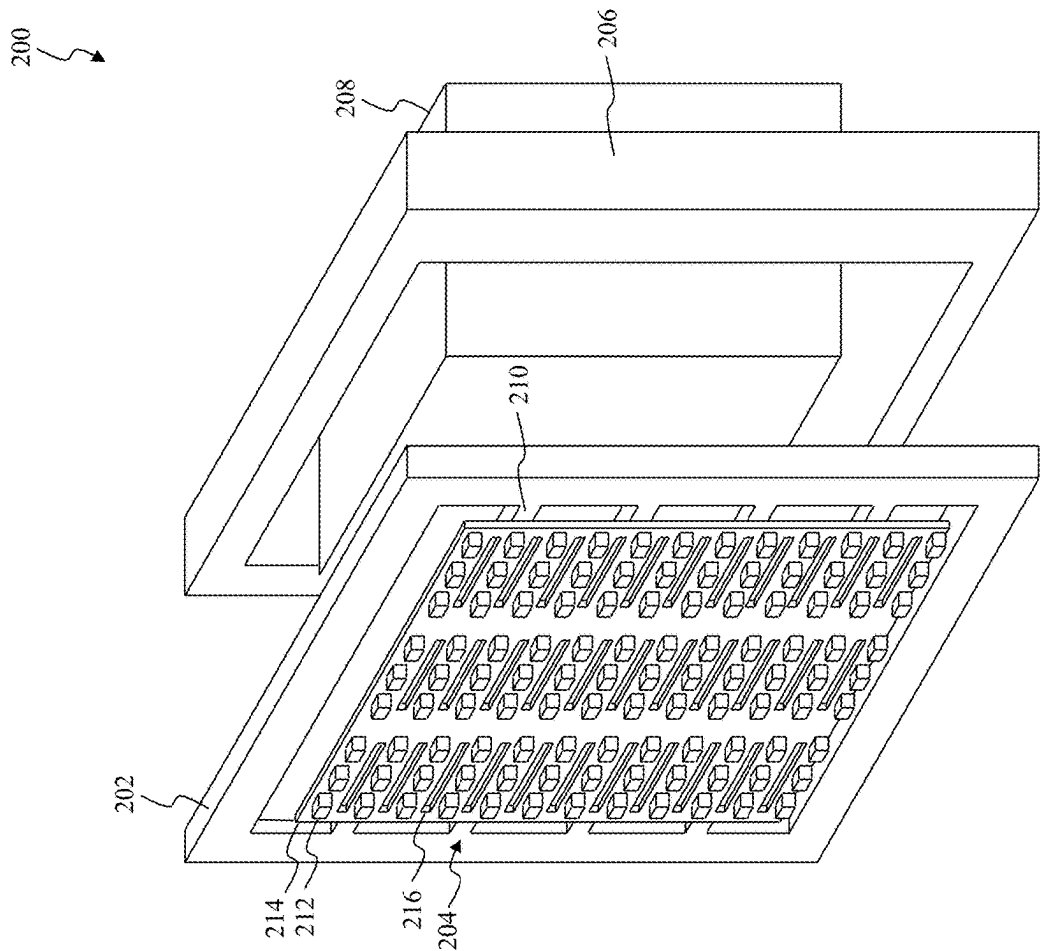


FIG. 2

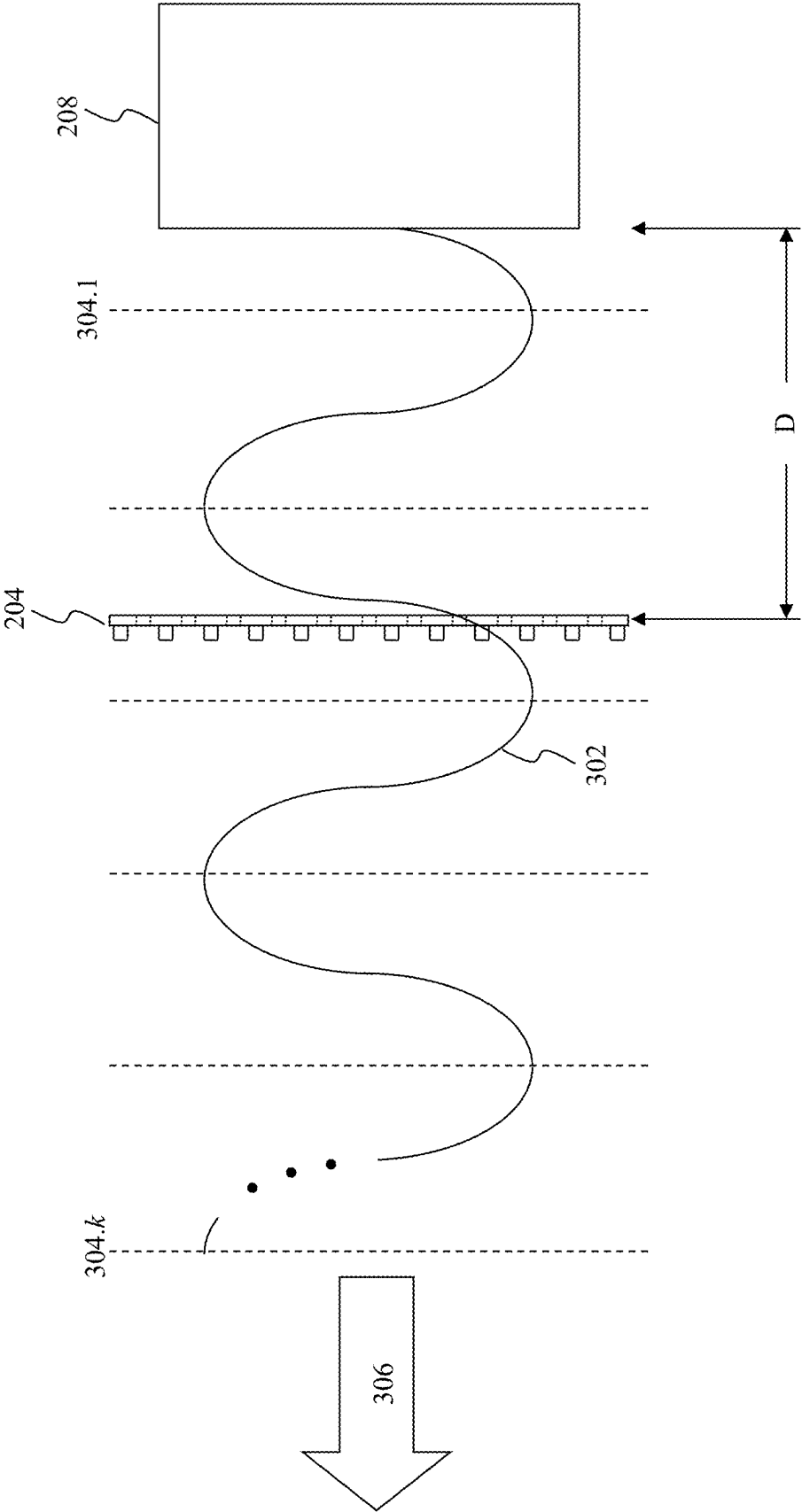


FIG. 3A

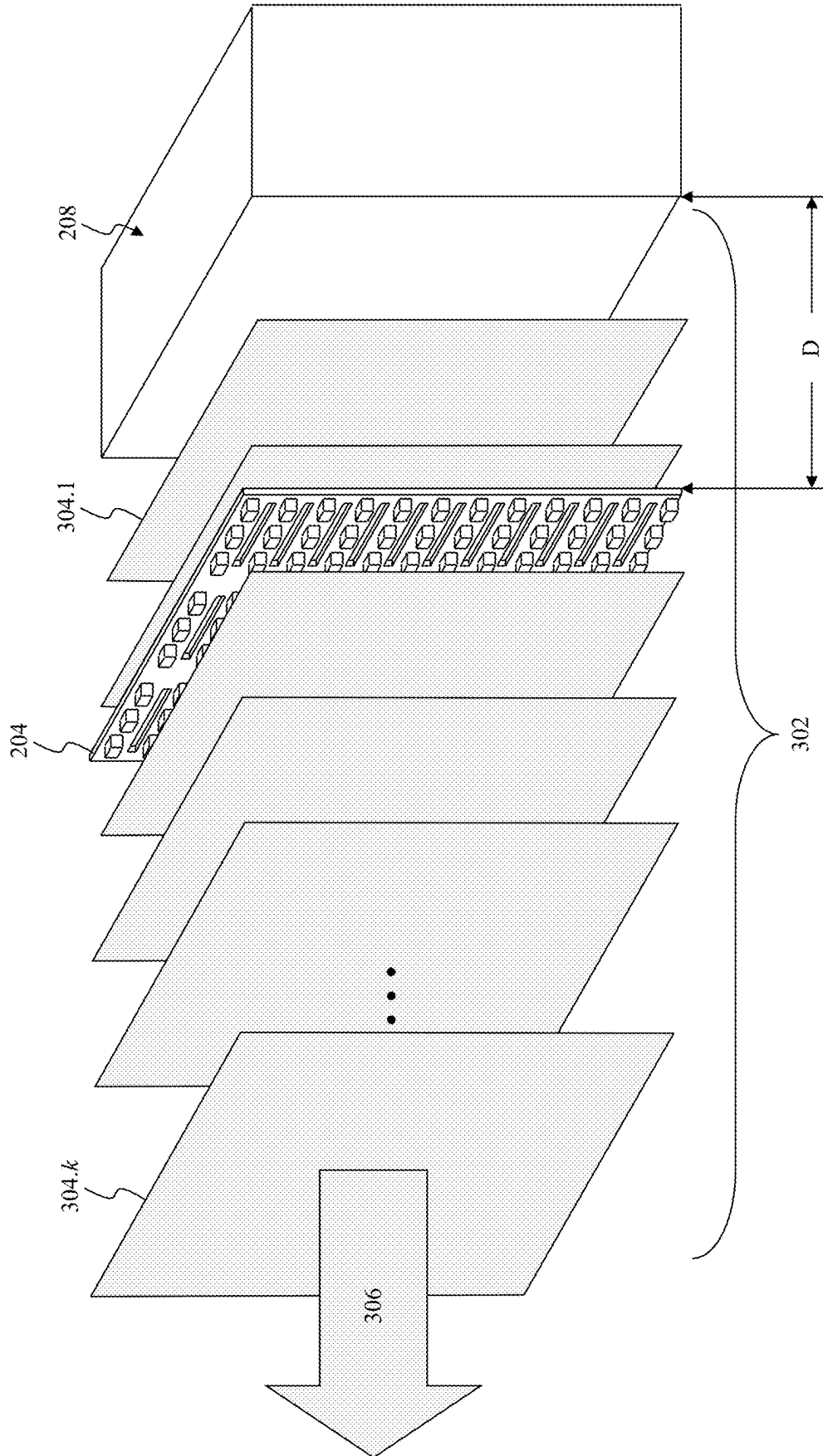


FIG. 3B

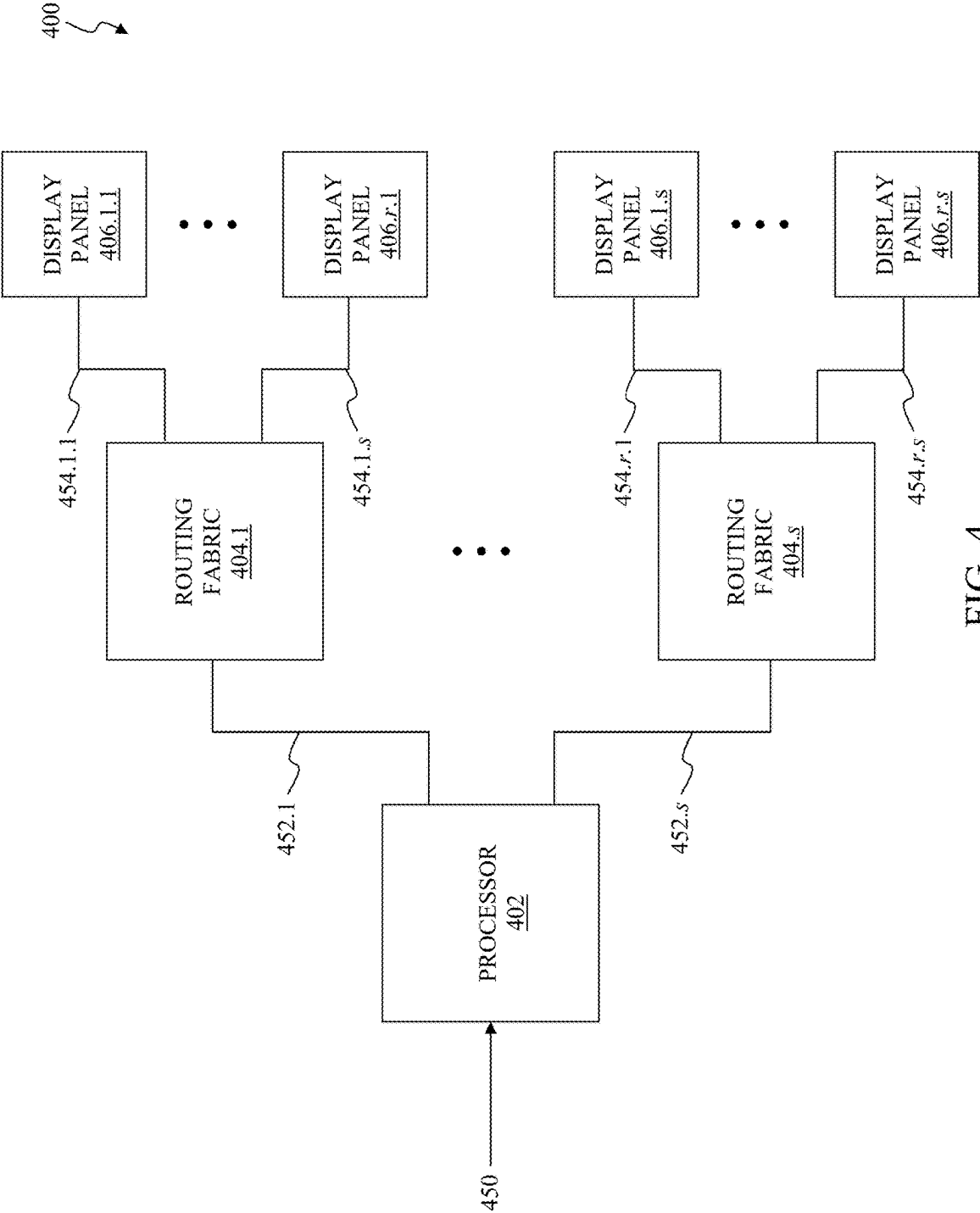


FIG. 4

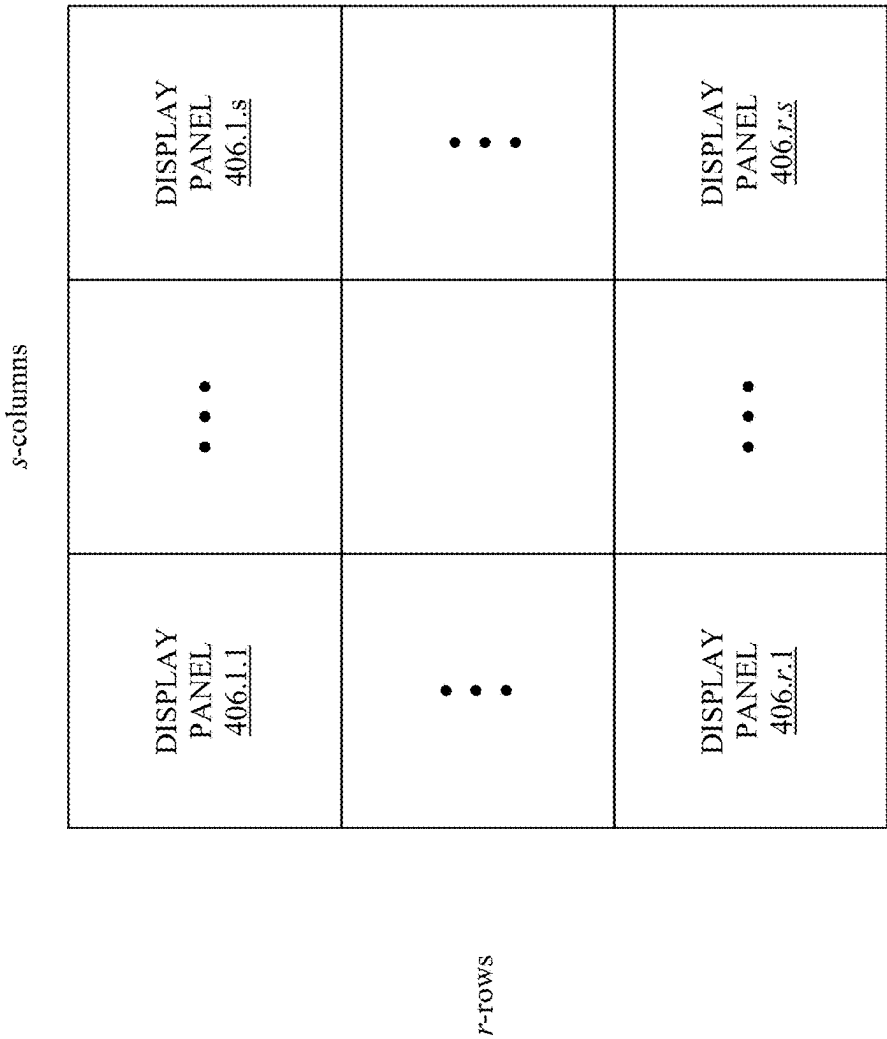


FIG. 5

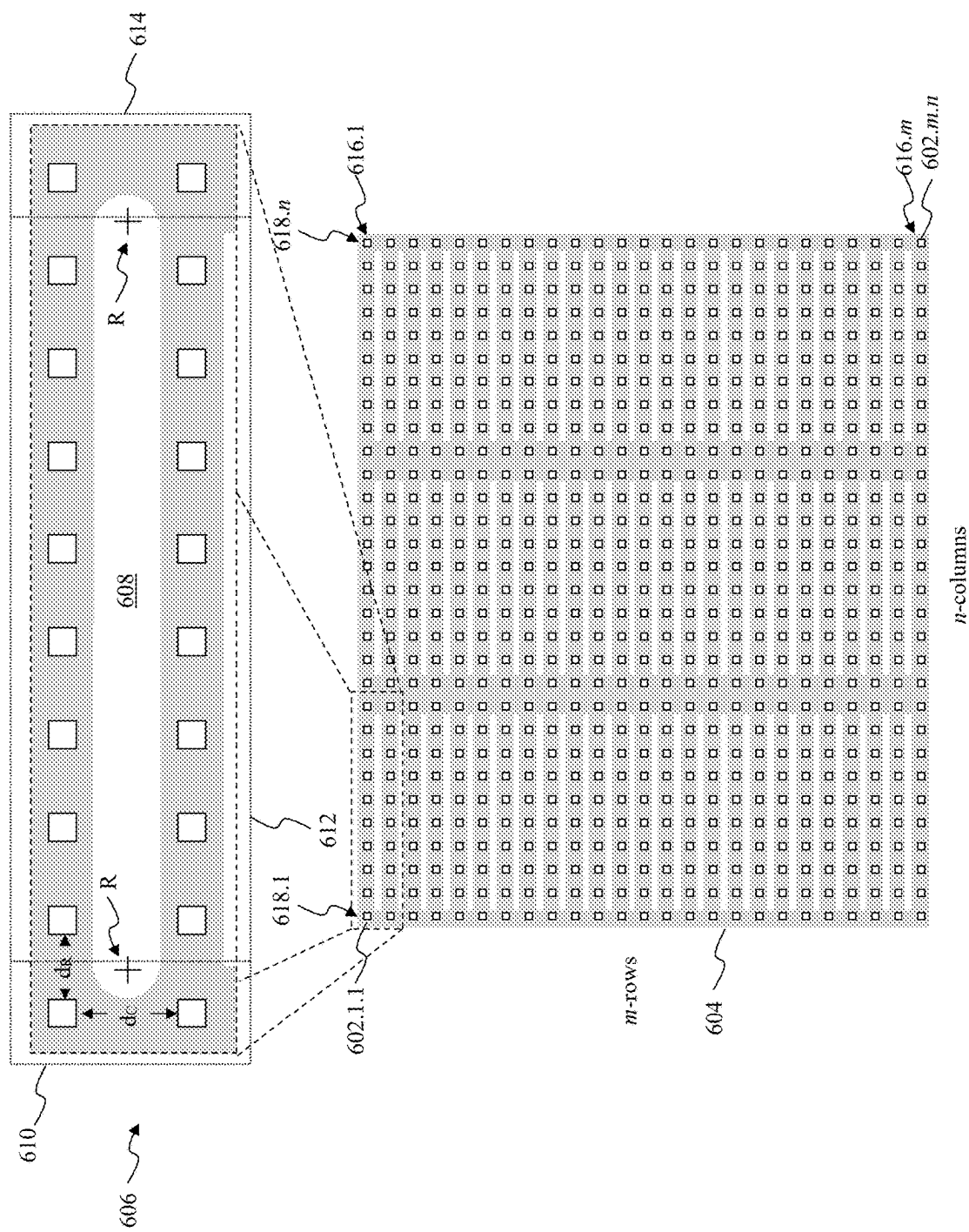


FIG. 6

INTEGRATED AUDIOVISUAL SYSTEM

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. patent application Ser. No. 18/521,021, filed on Nov. 28, 2023, which is a continuation of U.S. patent application Ser. No. 17/734,995, filed on May 2, 2022, now U.S. Pat. No. 11,375,305, now U.S. Pat. No. 11,856,346, which is a continuation of U.S. patent application Ser. No. 16/898,142, filed on Jun. 10, 2020, now U.S. Pat. No. 11,375,305, which claims the benefit of U.S. Provisional Application No. 62/859,930, filed on Jun. 11, 2019, each of which is incorporated herein by reference in its entirety.

BACKGROUND

[0002] The United States Media and Entertainment Industry is the largest in the world. The United States Media and Entertainment Industry represents a third of the global media and entertainment industry which delivers events, such as musical events, theatrical events, sporting events, and/or motion picture events, to spectators for their viewing pleasure. Presently, venues, such as music venues and/or sporting venues to provide an example, deliver these events to the spectators using conventional audio-visual systems having various display screens surrounded by loudspeakers in plain sight of the spectators. Not only do these conventional loudspeakers occupy space in the venues that could be used for displaying the event, venues often undergo various measures to cover these conventional loudspeakers from the field of view of the spectators.

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, features are not drawn to scale. In fact, the dimensions of the features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIG. 1 illustrates a pictorial representation of an exemplary venue according to an exemplary embodiment of the present disclosure;

[0005] FIG. 2 illustrates a pictorial representation of an exemplary integrated audio-visual system that can be implemented within the exemplary venue according to an exemplary embodiment of the present disclosure;

[0006] FIG. 3A and FIG. 3B illustrate simplified diagrams of exemplary operations of the exemplary integrated audio-visual system according to exemplary embodiments of the present disclosure;

[0007] FIG. 4 illustrates a block diagram of an exemplary visual display that can be implemented within the exemplary integrated audio-visual system according to an exemplary embodiment of the present disclosure;

[0008] FIG. 5 illustrates an exemplary arrangement of visual display panels that that can be implemented within the exemplary integrated audio-visual system according to an exemplary embodiment of the present disclosure; and

[0009] FIG. 6 illustrates an exemplary visual display panel that that can be implemented within the exemplary integrated audio-visual system according to an exemplary embodiment of the present disclosure.

DETAILED DESCRIPTION

[0010] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the examples. This repetition does not in itself dictate a relationship between the embodiments and/or configurations discussed.

Overview

[0011] An integrated audio-visual system is disclosed for delivering an event to one or more spectators. The integrated audio-visual system includes one or more loudspeakers that are positioned behind one or more visual displays to be effectively behind the one or more visual displays to be shielded from the field of view of the one or more spectators. The one or more visual displays are specially designed and manufactured to allow sound associated with the event to propagate from the one or more loudspeakers with minimal acoustical distortion and/or minimum acoustical vibration while presenting a visual representation of the event to the one or more spectators. Moreover, the one or more loudspeakers and the one or more visual displays are situated to be a predetermined displacement from each other to further minimize acoustical distortion.

Exemplary Venue

[0012] FIG. 1 illustrates a pictorial representation of an exemplary venue according to an exemplary embodiment of the present disclosure. In the exemplary embodiment illustrated in FIG. 1, a venue 100 represents a location for hosting an event. For example, the venue 100 can represent a music venue, for example, a music theater, a music club, and/or a concert hall, a sporting venue, for example, an arena, a convention center, and/or a stadium, and/or any other suitable venue that will be apparent to those skilled in the relevant(s) without departing the spirit and scope of the present disclosure. The event can represent a musical event, a theatrical event, a sporting event, a motion picture, and/or any other suitable event that will be apparent to those skilled in the relevant(s) without departing the spirit and scope of the present disclosure. As illustrated in FIG. 1, one or more spectators 102 experience visual and/or auditory representations of the event delivered from an integrated audio-visual system 104 within the venue 100. Although the discussion of the integrated audio-visual system 104, as well as other integrated audio-visual systems described herein, to follow describes these integrated audio-visual systems as delivering sound and visual representations of an event, those skilled in the relevant art(s) will recognize that the sound and the visual representation delivered by these integrated audio-visual systems need not be related to the same event without departing from the spirit and scope of the present disclosure. For example, these integrated audio-

visual systems can deliver sound of a musical song that is unrelated to a visual representation of a motion picture.

[0013] As to be described in further detail below, the integrated audio-visual system **104** delivers an audiovisual viewing experience of the event to the one or more spectators **102** through accurate and precise integration of its audiovisual components. For example, the audiovisual components within the integrated audio-visual system **104** can include one or more loudspeakers for delivering sound representing an audio component of the event to the one or more spectators **102**. Herein, the terms “loudspeaker” or “loudspeakers” refer to electrical, mechanical, and/or electro-mechanical devices which convert an electrical audio signal into a corresponding sound. The one or more loudspeakers can include one or more super tweeters, one or more tweeters, one or more mid-range speakers, one or more woofers, one or more subwoofers, one or more full-range speakers to provide some examples. Herein, the term “sound” refers to electrical waveforms in the general audible frequency whose frequency is in the band audible to the average human. Those skilled in the relevant art(s) will recognize the generally accepted standard range of audible frequencies for humans is between approximately 20 Hertz (Hz) and approximately 20 kilohertz (kHz). In some situations, at least some of these loudspeakers can be particularly arranged to form one or more line arrays of loudspeakers. In this example, the audiovisual components within the integrated audio-visual system **104** can further include one or more visual displays, such as one or more light-emitting diode (LED) displays, one or more organic light-emitting diode (OLED) displays, and/or one or more quantum dots (QDs) displays to provide some examples, for delivering a visual component of the event to the one or more spectators **102**.

[0014] The one or more visual displays, as to be further described in more detail below, are specially designed and manufactured to allow sound associated with the event to propagate from the one or more loudspeakers to the one or more spectators **102** with minimal acoustical distortion and/or minimum acoustical vibration while presenting the visual representation of the event to the one or more spectators **102**. For example, the one or more visual displays are particularly designed and manufactured onto one or more substrates, such as printed circuit board substrates or semiconductor substrates to provide some examples, having openings to allow the sound associated with the event to propagate from the one or more loudspeakers effectively passing through the one or more visual displays onto the one or more spectators **102**. Moreover, the one or more loudspeakers and the one or more visual displays are situated within the integrated audio-visual system **104** to be a predetermined displacement distance from each other. This predetermined displacement can allow for the formation of one or more wavefronts of the sound before the sound passes through the openings of the one or more visual displays to further minimize acoustical distortion.

Exemplary Integrated Audio-Visual System that can be Implemented within the Exemplary Venue

[0015] FIG. 2 illustrates a pictorial representation of an exemplary integrated audio-visual system that can be implemented within the exemplary venue according to an exemplary embodiment of the present disclosure. As to be described below, an integrated audio-visual system **200** is specially designed and manufactured to allow sound repre-

sending an audio component of an event to propagate through one or more visual displays while these visual displays are delivering a visual representation of the event. As illustrated in FIG. 2, the integrated audio-visual system **200** includes a visual display mechanical housing **202**, a visual display **204**, a loudspeaker mechanical housing **206**, and one or more loudspeakers **208**. The integrated audio-visual system **200** can represent an exemplary embodiment of the integrated audio-visual system **104** as described above in FIG. 1.

[0016] The visual display mechanical housing **202** represents a mechanical housing for securing the visual display **204**. The visual display mechanical housing **202** can be implemented using one or more metallic materials, such as iron, steel, copper, bronze, brass, aluminum, or magnesium to provide some examples, one or more non-metallic materials, such as wood, plastic, or glass, and/or any combination thereof. As illustrated in FIG. 2, the visual display mechanical housing **202** represents a frame or a housing for mechanically securing the visual display **204** from substantial acoustical vibration that can occur as the sound passes through the visual display **204**. As to be described in further detail below, the visual display **204** can include multiple visual display panels that functionally cooperate to deliver the visual representation of the event. In the exemplary embodiment illustrated in FIG. 2, the visual display mechanical housing **202** can include one or more mechanical cross members **210** for securing the multiple visual display panels to the visual display mechanical housing **202**. The one or more mechanical cross members **210** can be designed and manufactured to allow the sound associated with the event to propagate from the one or more loudspeakers **208** through the visual display mechanical housing **202** with minimal distortion and/or minimum acoustical vibration.

[0017] The visual display **204** delivers a visual component of the event. In the exemplary embodiment illustrated in FIG. 2, the visual display **204** includes multiple visual display panels that functionally cooperate to deliver the visual representation of the event. In this exemplary embodiment, these multiple visual display panels are mechanically attached to the visual display mechanical housing **202** using the one or more mechanical cross members **210**. For example, these multiple visual display panels can be arranged as a series of rows and a series of columns of visual display panels to form an array of visual display panels. In this example, rows of visual display panels can be mechanically attached to the mechanical cross members **210**. Moreover, the multiple visual display panels include visual emission elements **212**, such as light-emitting diodes (LEDs), organic light-emitting diode (OLEDs), and/or quantum dots (QDs) to provide some examples, to deliver the visual representation of the event. In an exemplary embodiment, the visual emission elements **212** can be arranged as a series of rows and a series of columns of visual emission elements to form an array of visual emission elements **212** to deliver the visual representation of the event as illustrated in FIG. 2. However, those skilled in the relevant art(s) will recognize other arrangements as possible for the visual emission elements **212** without departing from the spirit and scope of the present disclosure. As additionally illustrated in FIG. 2, the visual emission elements **212** can be situated onto a substrate **214**, such as a printed circuit board substrate or a semiconductor substrate to provide some examples. In an exemplary embodiment, the visual emission elements **212**

can be situated onto the substrate **214** using reflow soldering and/or wave soldering to provide some examples. In this exemplary embodiment, the substrate **214** can include an arrangement of pads, also referred to as footprints, to attach the visual emission elements **212** onto the substrate **214**. As further illustrated in FIG. 2, the substrate **214** can include one or more openings **216** to allow the sound associated with the event to propagate through the visual display **204** with minimal distortion and/or vibration as to be described in further detail below. The one or more openings **216** represent geometric regions within the visual display **204** in which the substrate **214** has been removed. In the exemplary embodiment illustrated in FIG. 2, the one or more openings **216** can be characterized as being rectangular geometric shapes as illustrated in FIG. 2, circular geometric shapes, or polygonal geometric shapes to provide some examples, and/or any suitable irregular geometric shapes, such as an irregular polygonal geometric shapes to provide an example are possible without departing from the spirit and scope of the present disclosure. It should be noted that the visual emission elements **212** and/or the number of the one or more openings **216** illustrated in FIG. 2 are for exemplary purposes only. Those skilled in the relevant art(s) will recognize the visual display **204** can include more or less visual emission elements **212** and/or the number of the one or more openings **216** than illustrated in FIG. 2 without departing from the spirit and scope of the present disclosure.

[0018] The loudspeaker mechanical housing **206** represents a mechanical housing for securing the one or more loudspeakers **208** to be effectively behind the visual display **204**. The loudspeaker mechanical housing **206** can be implemented using one or more metallic materials, such as iron, steel, copper, bronze, brass, aluminum, or magnesium to provide some examples, one or more non-metallic materials, such as wood, plastic, or glass, and/or any combination thereof. As illustrated in FIG. 2, the loudspeaker mechanical housing **206** represents a frame or a housing for mechanically securing the one or more loudspeakers **208**. In an exemplary embodiment, the loudspeaker mechanical housing **206** is designed and manufactured to mechanically dampen at least some of the acoustical vibration generated by the one or more loudspeakers **208** in providing the sound. In the exemplary embodiment illustrated in FIG. 2, the loudspeaker mechanical housing **206** mechanically positions the one or more loudspeakers **208** to be effectively behind the visual display **204** to be shielded from the field of view of the one or more spectators.

[0019] The one or more loudspeakers **208** deliver the sound representing an audio component of the event. The one or more loudspeakers **208** can include one or more super tweeters, one or more tweeters, one or more mid-range speakers, one or more woofers, one or more subwoofers, one or more full-range speakers, and/or any other suitable device that is capable of reproducing the audible frequency range, or a portion thereof, that will be apparent to those skilled in the relevant art(s) without departing from the spirit and scope of the present disclosure. In some situations, the one or more loudspeakers **208** can include multiple loudspeakers mounted in a theoretical line to form one or more line arrays of loudspeakers which are fed in phase. In some situations, loudspeakers within each line array of loudspeakers are of similar speaker type, for example, super tweeters, tweeters, mid-range speakers, woofers, subwoofers, or full-range speakers. In other situations, multiple line arrays of loud-

speakers can be stacked onto each other at slightly different vertical angles to cover a greater depth of field than a single line array of loudspeakers. As to be described in further detail below, the visual display **204** and the one or more loudspeakers **208** are situated to be a predetermined displacement from each other. This predetermined displacement allows for the formation of the sound by the one or more loudspeakers **208** before the sound passes through the openings of the visual display **204** to further minimize acoustical distortion.

Exemplary Operation of the Exemplary Integrated Audio-Visual System

[0020] FIG. 3A and FIG. 3B illustrate simplified diagrams of exemplary operations of the exemplary integrated audio-visual system according to exemplary embodiments of the present disclosure. As described above in FIG. 2, the one or more auditory speakers **208** generate an auditory waveform representing an audio component of an event which propagates through the visual display **204** while the visual display **204** is delivering a visual representation of the event. As illustrated in FIG. 3A, the one or more loudspeakers **208** generate sound **302** representing the audio component of the event which propagates through the visual display **204**. FIG. 3A illustrates an exemplary side view of the visual display **204** and the one or more loudspeakers **208**. Those skilled in the relevant art(s) will recognize the waveform **302** can differ from that depicted in FIG. 3A and FIG. 3B based upon different implementations of the visual display **204** and/or the one or more loudspeakers **208** without departing from the spirit and scope of the present disclosure. In the exemplary embodiment illustrated in FIG. 3A, the sound **302** represents a plane wave that can be characterized by planar wavefronts **304.1** through **304.k**. The planar wavefronts **304.1** through **304.k**, which represent surfaces of constant phase of the sound **302**, are further illustrated in FIG. 3B as being parallel planes normal to a phase velocity vector **306**.

[0021] In the exemplary embodiments illustrated in FIG. 3A and FIG. 3B, the visual display **204** can introduce acoustical distortion into the sound **302** as the sound **302** propagates through the openings of the visual display **204**. However, as illustrated in FIG. 3A and FIG. 3B and described above, the visual display **204** and the one or more loudspeakers **208** can be situated to be a predetermined displacement, denoted as displacement distance **D** in FIG. 3A and FIG. 3B, from each other to minimize this acoustical distortion. The displacement distance **D** allows for one or more of the planar wavefronts **304.1** through **304.k** to be formed before the sound **302** passes through the openings of the visual display **204** to minimize acoustical distortion of the sound **302** as the sound **302** propagates through the openings of the visual display **204**. In an exemplary embodiment, the displacement distance **D** is approximately 50 centimeters (cm).

Exemplary Visual Display that can be Implemented within the Exemplary Integrated Audio-Visual System

[0022] FIG. 4 illustrates a block diagram of an exemplary visual display that can be implemented within the exemplary integrated audio-visual system according to an exemplary embodiment of the present disclosure. A visual display system **400** delivers a visual component of an event to one or more spectators as described above. In the exemplary embodiment illustrated in FIG. 4, the visual display system **400** includes a processor **402**, routing fabric **404.1** through

404.s, and visual display panels 406.1.1 through 406.r.s. The visual display system 400 can represent an exemplary embodiment of the visual display 204 as described above in FIG. 2.

[0023] The processor 402 controls overall configuration and/or operation of the visual display system 400. In an exemplary embodiment, the processor 402 can be situated remote from the routing fabric 404.1 through 404.s and the visual display panels 406.1.1 through 406.r.s. For example, the visual display system 400 can be located within a venue, such as the venue 100 as described above. In this example embodiment, the processor 402 can be implemented using a series of one or more processors that can be mounted in one or more equipment racks in a different location, for example, a different room, within the venue than the visual display panels 406.1.1 through 406.r.s. In the exemplary embodiment illustrated in FIG. 4, the processor 402 receives digital video data 450 representing the visual component of the event to deliver to the one or more spectators. Preferably, the event can be pre-recorded onto stored on a machine-readable medium, which can be read and executed by one or more processors. A machine-readable medium can include any mechanism for storing or transmitting information in a form readable by a machine (e.g., a computing circuitry). For example, a machine-readable medium can include non-transitory machine-readable mediums such as read only memory (ROM); random access memory (RAM); magnetic disk storage media; optical storage media; flash memory devices; and others. As another example, the machine-readable medium can include transitory machine-readable medium such as electrical, optical, acoustical, or other forms of propagated signals (e.g., carrier waves, infrared signals, digital signals, etc.). Thereafter, a media distribution system can provide the event is stored on the machine-readable medium to the processor as the digital video data 450. This media distribution system can be within a venue, such as the venue 100, or outside of the venue with capabilities to service multiple venues. Alternatively, the digital video data 450 representing the event can be streamed to the processor 402 in real-time, or near real-time, as the event is occurring. In an exemplary embodiment, the processor 402 can supplement the digital video data 450 with supplemental digital data, such as virtual advertising and graphic effects, prior to delivering the digital video data 450 to the visual display panels 406.1.1 through 406.r.s.

[0024] The processor 402 processes the digital video data 450 to provide the control signals 452.1 through 452.s. In the exemplary embodiment illustrated in FIG. 4, each control signal from among the control signals 452.1 through 452.s configures a corresponding group of visual display panels from among the visual display panels 406.1.1 through 406.r.s, for example, a corresponding column of visual display panels, to deliver their corresponding portion of the visual component of the event to one or more spectators. For example, the processor 402 can map, referred to as pixel map, the digital video data 450 from an image space into a display space for display onto the visual display panels 406.1.1 through 406.r.s. In this example, the pixels of the digital video data 450 can be assigned to various coordinates in the image space and the visual display panels 406.1.1 through 406.r.s can be assigned to various coordinates in the display space. In this example, the processor 402 translates or maps the pixels of the digital video data 450 from the image space to the display space. Thereafter, the processor

402 provides the control signals 452.1 through 452.s to cause the display panels 406.1.1 through 406.r.s to display the digital video data 450 in the display space. In an exemplary embodiment, the control signals 452.1 through 452.s can represent serial control signals with these serial control signals corresponding to groups of visual display panels from among the visual display panels 406.1.1 through 406.r.s, for example, corresponding columns of visual display panels. Alternatively, or in addition to, the control signals 452.1 through 452.s can represent parallel control signals with these parallel control signals corresponding to individual visual display panels from among the visual display panels 406.1.1 through 406.r.s. In some situations, the digital video data 450 can be encoded in accordance with a video coding format, such as a version of a Moving Picture Experts Group (MPEG) video coding format or a version of an AOMedia Video 1 (AV1) video coding format to provide some examples. In these situations, the processor 402 decodes the digital video data 450 in accordance with the format before processing the digital video data 450, as to be described in further detail below, for delivery to the visual display panels 406.1.1 through 406.r.s. Optionally, the processor 402 can perform motion compensation, inverse discrete cosine transforms (iDCTs), inverse modified discrete cosine transforms (iMDCTs), in-loop deblocking filters, intra-frame predictions, inverse quantizations (IQs), variable-length decoding (VLD), spatial-temporal deinterlacing, automatic interlace/progressive source detection, and/or bit-stream processing and perfect pixel positioning to manipulate the images, or the series or images, of the event within the digital video data 450.

[0025] The routing fabric 404.1 through 404.s selectively routes the control signals 452.1 through 452.s for delivery to the visual display panels 406.1.1 through 406.r.s as control signals 454.1.1 through 454.r.s. In some situations, the processor 402 can be communicatively coupled to the routing fabric 404.1 through 404.s using suitable communication mediums, such as one or more Ethernet networking cables, such as one or more category 5 (Cat 5) or category 6 (Cat 6) Ethernet networking cables to provide some examples, one or more optical fiber cables, and/or one or more copper core communication cables, such as one or more coaxial cables, one or more ribbon cables, one or more shielded cables, and/or one or more twinax cables to provide some examples. In the exemplary embodiment illustrated in FIG. 4, the routing fabric 404.1 through 404.s can convert the control signals 452.1 through 452.s from serial control signals to provide the as control signals 454.1.1 through 454.r.s as parallel control signals for delivery to the visual display panels 406.1.1 through 406.r.s. As to be described in further detail below, the visual display panels 406.1.1 through 406.r.s can be arranged as a series of r rows and a series of s columns of visual display panels to form an array of visual display panels.

[0026] The visual display panels 406.1.1 through 406.r.s utilize the control signals 454.1.1 through 454.r.s to collectively deliver the visual component of the event to one or more spectators. In an exemplary embodiment, the visual display panels 406.1.1 through 406.r.s can be arranged as a series of r rows and a series of s columns of visual display panels to form an array of visual display panels. In the exemplary embodiment illustrated in FIG. 4, the visual display panels 406.1.1 through 406.r.s utilize the control signals 454.1.1 through 454.r.s to control operation of

various visual emission elements of the visual display panels **406.1.1** through **406.r.s** to collectively deliver the visual component of the event to one or more spectators. In this exemplary embodiment, the control signals **454.1.1** through **454.r.s** can represent power signals, ground signals, clocking signals, and/or any other suitable signal that can be used to control the operation of the various visual emission elements that will be apparent to those skilled in the relevant art(s) without departing from the spirit and scope of the present disclosure. Moreover, the various visual emission elements can include light-emitting diodes (LEDs), organic light-emitting diode (OLEDs), quantum dots (QDs), and/or any combination thereof to deliver the visual representation of the event. The control signals **454.1.1** through **454.r.s** can be utilized to activate, or turn-on, these various visual emission elements to emit light with the color and/or intensity of the light being dependent upon chemical properties of the various visual emission elements. Alternatively, or in addition to, the control signals **454.1.1** through **454.r.s** can be utilized to deactivate, or turn-off, these various visual emission elements to not emit light. For example, the control signals **454.1.1** through **454.r.s** can be configured to forward bias the various visual emission elements to activate these visual emission elements or to reverse bias the various visual emission elements to deactivate these visual emission elements.

Exemplary Arrangement of Visual Display Panels that can be Implemented within the Exemplary Integrated Audio-Visual System

[0027] FIG. 5 illustrates an exemplary arrangement of visual display panels that that can be implemented within the exemplary integrated audio-visual system according to an exemplary embodiment of the present disclosure. As described above in FIG. 4, the visual display panels **406.1.1** through **406.r.s** can be arranged as the series of r rows and the series of s columns of visual display panels to form the array of visual display panels. FIG. 5 illustrates such an arrangement of the visual display panels **406.1.1** through **406.r.s** in the series of r rows and the series of s columns. Exemplary Visual Display Panel that can be Implemented within the Exemplary Integrated Audio-Visual System

[0028] FIG. 6 illustrates an exemplary visual display panel that that can be implemented within the exemplary integrated audio-visual system according to an exemplary embodiment of the present disclosure. In the exemplary embodiment illustrated in FIG. 6, a visual display panel **600** can be specially designed and manufactured to allow sound, such as the sound **302** as described above in FIG. 3, of an event to propagate from one or more loudspeakers to one or more spectators of the event with minimal acoustical distortion and/or minimum acoustical vibration while presenting a visual representation of the event to the one or more spectators. As to be described below, the visual display panel **600** is particularly designed and manufactured onto a printed circuit board substrate having openings to allow the sound associated with the event to propagate from the one or more loudspeakers effectively passing through the visual display panel **600** onto the one or more spectators. As illustrated in FIG. 6, the visual display panel **600** includes visual emission elements **602.1.1** through **602.m.n** that are situated onto a printed circuit board substrate **604**.

[0029] In the exemplary embodiment illustrated in FIG. 6, the visual emission elements **602.1.1** through **602.m.n** can be arranged in a series of m rows and a series of n columns onto

the printed circuit board substrate **604** to form a rectangular array of visual emission elements. However, those skilled in the relevant art(s) will recognize other arrangements for the visual emission elements **602.1.1** through **602.m.n** are possible without departing from the spirit and scope of the present disclosure. For example, the visual emission elements **602.1.1** through **602.m.n** and/or the printed circuit board substrate **604** can be any suitable regular geometric shape, such as a rectangular geometric shape as illustrated in FIG. 6, a circular geometric shape, or a polygonal geometric shape to provide some examples, and/or any suitable irregular geometric shape, such as an irregular polygonal geometric shape to provide an example are possible without departing from the spirit and scope of the present disclosure. Furthermore, although the visual emission elements **602.1.1** through **602.m.n** and the printed circuit board substrate **604** have similar geometric arrangement in FIG. 6, those skilled in the relevant art(s) will recognize the geometric arrangements of the visual emission elements **602.1.1** through **602.m.n** and the printed circuit board substrate **604** can differ without departing from the spirit and scope of the present disclosure. In an exemplary embodiment, the visual emission elements **602.1.1** through **602.m.n** can be situated onto the printed circuit board substrate **604** using reflow soldering and/or wave soldering to provide some examples. The visual emission elements **602.1.1** through **602.m.n** can be implemented using light-emitting diodes (LEDs), organic light-emitting diodes (OLEDs), and/or one or more quantum dots (QDs) to provide some examples.

[0030] Moreover, visual emission elements **602.1.1** through **602.m.n** from each row from among the series of m rows are spaced apart from each other by an approximate row distance d_R and visual emission elements **602.1.1** through **602.m.n** from each column from among the series of n columns are spaced apart from each other by an approximate column distance d_C as illustrated in an exploded view **606** of the visual display panel **600** in FIG. 6. In an exemplary embodiment, the row distance d_R and the column distance d_C are equidistant, for example, approximately six (6) millimeters (mm) or approximately nine (9) mm. However, those skilled in the relevant art(s) will recognize that the spacing between the visual emission elements **602.1.1** through **602.m.n** can be different than as depicted in FIG. 6 without departing from the spirit and scope of the present disclosure. The row distance d_R and the column distance d_C are selectively chosen to provide a relatively smooth viewing experience of the visual representation of the event to the one or more spectators while taking into consideration the acoustical distortion of the sound associated with the event as it propagates through the visual display panel **600** to the one or more spectators. As to be described in further detail below, the visual display panel **600** includes openings **608** to allow the sound associated with the event to propagate from the one or more loudspeakers effectively passing through the visual display panel **600** onto the one or more spectators. And as illustrated in FIG. 6, sizes and/or shapes of the openings **608** are bound by the row distance d_R and the column distance d_C with greater row distances d_R and/or greater column distances d_C allowing for larger openings **608**. However, if the row distances d_R and/or the column distances d_C are too large, the visual representation of the event can appear pixelated to the one or more spectators. On the other hand, if the row distances d_R and/or the column distances d_C are too small, the visual display panel **600** can

cause significant acoustical distortion of the sound associated with the event as the sound propagates through the visual display panel 600.

[0031] As described above, the visual display panel 600 includes the openings 608 to allow the sound associated with the event to propagate from the one or more loudspeakers to the one or more spectators. As illustrated in FIG. 6, the openings 608 represent geometric regions within the visual display panel 600 in which the printed circuit board substrate 604 has been removed. In the exemplary embodiment illustrated in FIG. 6, the openings 608 can be characterized as being rectangular geometric shapes as illustrated in FIG. 6, circular geometric shapes, or polygonal geometric shapes to provide some examples, and/or any suitable irregular geometric shapes, such as an irregular polygonal geometric shapes to provide an example as possible without departing from the spirit and scope of the present disclosure. Optionally, the openings 608 can be radially shaped in accordance with a radius R as illustrated in FIG. 6. This optional radial shaping effectively shapes transitions between edges of the openings 608 to further minimize distortion of the sound as the sound propagates through the visual display panel 600.

[0032] In the exemplary embodiment illustrated in FIG. 6, the openings 608 traverse between columns from among the series of n columns of visual emission elements 602.1.1 through 602.m.n. As illustrated in the exploded view 606, the openings 608 traverse from a first set 610 of visual emission elements from among the visual emission elements 602.1.1 through 602.m.n along a first column from among the n columns through a second set 612 of visual emission elements from among the visual emission elements 602.1.1 through 602.m.n to a third set 614 of visual emission elements from among the visual emission elements 602.1.1 through 602.m.n along a second column from among the n columns. This illustrated arrangement of the openings 608 in the exploded view 606 allows control signals, such as the control signals 454.1.1 through 454.r.s to provide an example, to be routed through the series of m rows and the series of n columns to control the operation of the visual emission elements 602.1.1 through 602.m.n. As illustrated in FIG. 6, the visual display panel 600 includes transmission lines 616.1 through 616.m and transmission lines 618.1 through 618.n to route these control signals through the series of m rows and the series of n columns to control the operation of the visual emission elements 602.1.1 through 602.m.n. The transmission lines 616.1 through 616.m and transmission lines 618.1 through 618.n can be implemented using planar waveguides, such as stripline or microstrip to provide some examples. In the exemplary embodiment illustrated in FIG. 6, the transmission lines 616.1 through 616.m are situated along rows from among the series of m rows to route these control signals to visual emission elements 602.1.1 through 602.m.n along these rows and the transmission lines 618.1 through 618.n are situated along rows from columns the series of n columns to route these control signals to rows along these columns. Alternatively, or in addition to, the transmission lines 618.1 through 618.n route these control signals to visual emission elements 602.1.1 through 602.m.n along these columns and the transmission lines 616.1 through 616.m route these control signals to columns along these rows.

CONCLUSION

[0033] The Detailed Description referred to accompanying figures to illustrate exemplary embodiments consistent with the disclosure. References in the disclosure to “an exemplary embodiment” indicates that the exemplary embodiment described can include a particular feature, structure, or characteristic, but every exemplary embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same exemplary embodiment. Further, any feature, structure, or characteristic described in connection with an exemplary embodiment can be included, independently or in any combination, with features, structures, or characteristics of other exemplary embodiments whether or not explicitly described.

[0034] The Detailed Description is not meant to limiting. Rather, the scope of the disclosure is defined only in accordance with the following claims and their equivalents. It is to be appreciated that the Detailed Description section, and not the Abstract section, is intended to be used to interpret the claims. The Abstract section can set forth one or more, but not all exemplary embodiments, of the disclosure, and thus, are not intended to limit the disclosure and the following claims and their equivalents in any way.

[0035] The exemplary embodiments described within the disclosure have been provided for illustrative purposes and are not intended to be limiting. Other exemplary embodiments are possible, and modifications can be made to the exemplary embodiments while remaining within the spirit and scope of the disclosure. The disclosure has been described with the aid of functional building blocks illustrating the implementation of specified functions and relationships thereof. The boundaries of these functional building blocks have been arbitrarily defined herein for the convenience of the description. Alternate boundaries can be defined so long as the specified functions and relationships thereof are appropriately performed.

[0036] Embodiments of the disclosure can be implemented in hardware, firmware, software application, or any combination thereof. Embodiments of the disclosure can also be implemented as instructions stored on a machine-readable medium, which can be read and executed by one or more processors. A machine-readable medium can include any mechanism for storing or transmitting information in a form readable by a machine (e.g., a computing circuitry). For example, a machine-readable medium can include non-transitory machine-readable mediums such as read only memory (ROM); random access memory (RAM); magnetic disk storage media; optical storage media; flash memory devices; and others. As another example, the machine-readable medium can include transitory machine-readable medium such as electrical, optical, acoustical, or other forms of propagated signals (e.g., carrier waves, infrared signals, digital signals, etc.). Further, firmware, software application, routines, instructions can be described herein as performing certain actions. However, it should be appreciated that such descriptions are merely for convenience and that such actions in fact result from computing devices, processors, controllers, or other devices executing the firmware, software application, routines, instructions, etc.

[0037] The Detailed Description of the exemplary embodiments fully revealed the general nature of the disclosure that others can, by applying knowledge of those skilled in relevant art(s), readily modify and/or adapt for

various applications such exemplary embodiments, without undue experimentation, without departing from the spirit and scope of the disclosure. Therefore, such adaptations and modifications are intended to be within the meaning and plurality of equivalents of the exemplary embodiments based upon the teaching and guidance presented herein. It is to be understood that the phraseology or terminology herein is for the purpose of description and not of limitation, such that the terminology or phraseology of the present specification is to be interpreted by those skilled in relevant art(s) in light of the teachings herein.

What is claimed is:

1. A visual display panel for a visual display, the visual display panel comprising:

a substrate having a plurality of openings;
a plurality of visual emission elements situated on the substrate and configured to deliver one or more images; wherein the plurality of openings are configured to allow one or more sounds generated by a loudspeaker within the venue to propagate through the substrate; and wherein the substrate is positioned at a displacement distance from the loudspeaker such that one or more wavefronts of the one or more sounds are formed prior to propagating through the plurality of openings.

2. The visual display panel of claim 1, wherein the plurality of visual emission elements are arranged in a series of rows and a series of columns to form an array of visual emission elements,

wherein the plurality of openings are positioned along one or more corresponding rows from among the series of rows, and

wherein the plurality of openings are positioned between two corresponding columns from among the series of columns.

3. The visual display panel of claim 1, wherein the plurality of visual emission elements comprise one or more light-emitting diodes (LEDs), organic light-emitting diodes (OLEDs), or quantum dots (QDs).

4. The visual display panel of claim 1, wherein the displacement distance is configured to allow the one or more sounds travel from the loudspeaker such that one or more surfaces of constant phase are established before the one or more sounds the plurality of openings.

5. The visual display panel of claim 3, wherein the constant phase comprises a constant phase relationship across one or more points on the one or more surfaces.

6. The visual display panel of claim 1, wherein plurality of openings and the displacement distance are configured to optimize the propagation of the one or more sounds through the plurality of openings such that the plurality of openings are shaped and arranged to optimize acoustical distortion and vibration as the one or more sounds pass through the openings.

7. The visual display panel of claim 1, wherein the displacement distance is 50 centimeters (cm).

8. A method for operating a display panel within a venue, the method comprising:

installing the display panel within the venue, the display panel including a substrate with a plurality of openings and a plurality of visual emission elements;

configuring a plurality of visual emission elements to deliver one or more images;

positioning the display panel at a displacement distance from a loudspeaker within the venue; and

propagating one or more wavefronts of one or more sounds generated by the loudspeaker through the plurality of openings.

9. The method of claim 8, further comprising:

arranging a plurality of visual emission elements in a series of rows and a series of columns to form an array of visual emission elements;

positioning the plurality of openings along one or more corresponding rows from among the series of rows; and positioning the plurality of openings between two corresponding columns from among the series of columns.

10. The method of claim 8, wherein the configuring comprises configuring one or more light-emitting diodes (LEDs), organic light-emitting diodes (OLEDs), or quantum dots (QDs) to display the one or more images.

11. The method of claim 8, wherein the positioning comprises positioning the display panel at the displacement distance to allow the one or more sounds to travel from the loudspeaker such that one or more surfaces of constant phase are established before the one or more sounds propagate through the plurality of openings.

12. The method of claim 11, wherein the positioning further comprises positioning the display panel at the displacement distance to allow the one or more sounds to travel from the loudspeaker such that one or more surfaces of a constant phase relationship across one or more points on the one or more surfaces are established before the one or more sounds propagate through the plurality of openings.

13. The method of claim 8, wherein the positioning the display panel at the displacement distance to optimize the propagation of the one or more sounds through the plurality of openings such that the plurality of openings are shaped and arranged to optimize acoustical distortion and vibration as the one or more sounds pass through the openings.

14. The method of claim 8, wherein the displacement distance is 50 centimeters (cm).

15. A visual display for use within a venue, the visual display comprising:

a plurality of visual display panels configured and arranged as a first series of rows and a first series of columns to form the visual display

each visual display panel from among the plurality of visual display panels comprising:

a substrate having a plurality of openings;

a plurality of visual emission elements situated on the substrate and configured to deliver one or more images;

wherein the plurality of openings are configured to allow one or more sounds generated by a loudspeaker within the venue to propagate through the substrate; and

wherein the substrate is positioned at a displacement distance from the loudspeaker such that one or more wavefronts of the one or more sounds are formed prior to propagating through the plurality of openings.

16. The visual display of claim 15, wherein the plurality of visual emission elements are arranged in a second series of rows and a second series of columns to form an array of visual emission elements,

wherein the plurality of openings are positioned along one or more corresponding rows from among the series of rows, and

wherein the plurality of openings are positioned between two corresponding columns from among the series of columns.

17. The visual display of claim **15**, wherein the plurality of visual emission elements comprise one or more light-emitting diodes (LEDs), organic light-emitting diodes (OLEDs), or quantum dots (QDs).

18. The visual display of claim **15**, wherein the displacement distance is configured to allow the one or more sounds travel from the loudspeaker such that one or more surfaces of constant phase are established before the one or more sounds the plurality of openings.

19. The visual display of claim **18**, wherein the constant phase comprises a constant phase relationship across one or more points on the one or more surfaces.

20. The visual display of claim **15**, wherein plurality of openings and the displacement distance are configured to optimize the propagation of the one or more sounds through the plurality of openings such that the plurality of openings are shaped and arranged to optimize acoustical distortion and vibration as the one or more sounds pass through the openings.

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